

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied Business Analytics Practical

Module - II

Practical – VII

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Class:TYIT	Batch:2
Date of Assignment: 3/10/26	Date/Time of Submission: 3/10/26

7. Sampling and Sampling Distributions:

a. Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.

CODE:

```
import pandas as pd
import numpy as np
import re
import matplotlib.pyplot as plt

# 1. Load the survey data
df = pd.read_csv("A Survey on Daily Study Hrs.csv")

# 2. Display first five records
print("First Five Records:")
print(df.head())

# 3. Study hours column
col = df.columns[4]
# Convert study-hour responses into numerical values
def convert_hours(value):
    value = str(value)
    # Convert ranges into midpoint
    match = re.search(r'(\d+(?:\.\d+)?)s*-\s*(\d+(?:\.\d+)?)',
value)
    if match:
        a = float(match.group(1))
        b = float(match.group(2))
        return (a + b) / 2
    # Convert single numerical values
    match = re.search(r'\d+(?:\.\d+)?', value)
    if match:
        return float(match.group())
    return np.nan

df["Study Hours"] = df[col].apply(convert_hours)

# Remove missing values
study_hours = df["Study Hours"].dropna()

# 4. Display collected data
print("\nTotal Responses:", len(df))

# 5. Population and sample
print("\nPopulation: TYIT students")
print("Sample: Students who responded to the survey")

# 6. Sampling technique
print("Sampling Technique: Simple Random Sampling")

# 7. Numerical variables
print("\nNumerical Variables:")
print("Age, Study Hours")

# 8. Categorical variables
print("\nCategorical Variables:")
print("Gender, Days per Week, Study Schedule, Study
Location")

# 9 & 10. Calculate statistics
sample_size = len(study_hours)
sample_mean = study_hours.mean()
sample_median = study_hours.median()
sample_std = study_hours.std()
print("\nStudy Hours Statistics:")
print("Sample Size:", sample_size)
print("Mean:", round(sample_mean, 2), "hours")
print("Median:", round(sample_median, 2), "hours")
print("Standard Deviation:", round(sample_std, 2), "hours")
# Display study hours in compact format
print("\nStudy Hours Data:")
for i in range(0, len(study_hours), 10):
    print(" | ".join(
        f'{j+1}:{study_hours.iloc[j]:.1f}'
        for j in range(i, min(i+10, len(study_hours)))) )

# 11. Chart
plt.hist(study_hours, bins=8, edgecolor="black")
plt.xlabel("Daily Study Hours")
plt.ylabel("Number of Students")
plt.title("Daily Study Hours of TYIT Students : Priti Yadav
T054")
plt.show()
```

OUTPUT:

```
First Five Records:
   Timestamp      1. Name  2. Gender  3. Age  \
0  9/17/2026 19:06:57  Priti Yadav   Female   19
1  9/17/2026 19:36:36      Sapna   Female   18
2  9/17/2026 19:37:30   Shreeom    Male   20
3  9/17/2026 19:38:33   Sangeeta   Female   29
4  9/17/2026 19:39:28    Shruti   Female   17

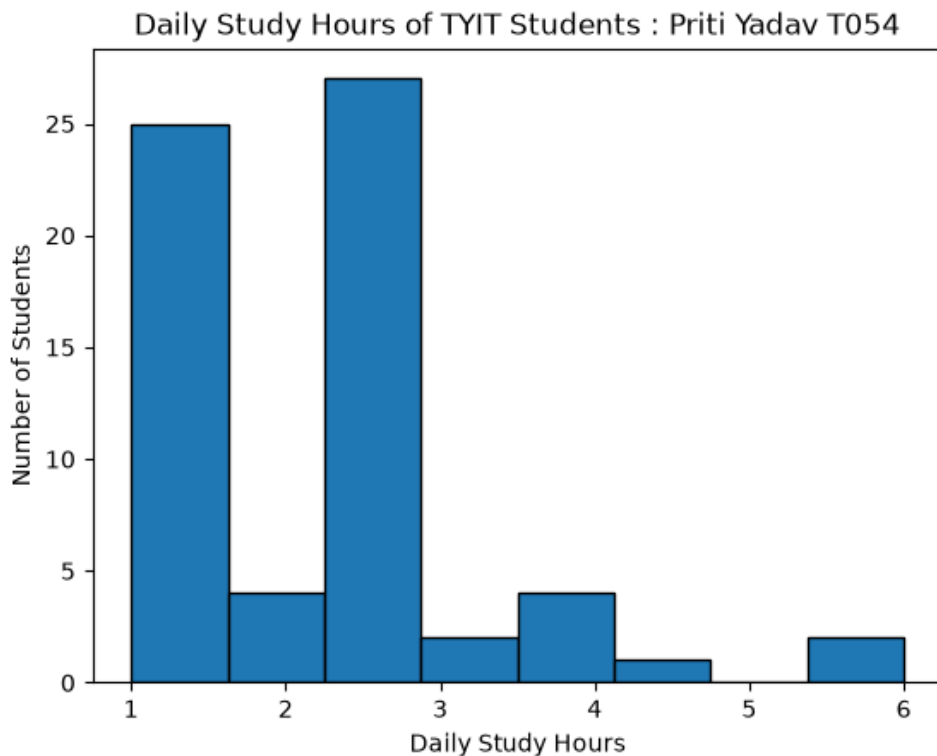
4. On average, how many hours do you study per day?  \
0                                     2-3 hours
1                                     4-5 hours
2                                     1-2 hours
3                                     2 hours
4                                     3
```

```

5. How many days per week do you study? \
0      3-4 days
1      3-4 days
2      5-6 days
3      1-2 days
4      7 days

6. Do you follow a fixed study schedule? \
0      Sometimes
1      No
...
31:2.5 | 32:2.5 | 33:2.5 | 34:1.5 | 35:1.5 | 36:1.5 | 37:1.5 | 38:1.5 | 39:1.5 | 40:1.5
41:1.5 | 42:1.5 | 43:2.5 | 44:1.5 | 45:1.5 | 46:1.5 | 47:1.5 | 48:1.5 | 49:1.5 | 50:2.0
51:6.0 | 52:6.0 | 53:4.0 | 54:1.0 | 55:4.0 | 56:1.5 | 57:1.5 | 58:1.5 | 59:1.5 | 60:2.5
61:2.5 | 62:2.5 | 63:2.5 | 64:2.5 | 65:2.5

```



Observations:

1. The average daily study time is about 2.3 hours.
2. The median study time is 2.5 hours per day.

b. Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters.

CODE:

```

# Select numerical variable
data = study_hours.to_numpy()
# 14. Select sample size
sample_size = 30
# 16. Number of repetitions
num_samples = 100
# Set random seed
np.random.seed(42)
# 17. Store sample means
sample_means = []
for i in range(num_samples):
    # Random sample of 30 observations
    sample = np.random.choice(
        data,
        size=sample_size,
        replace=False )
    # Calculate sample mean
    sample_mean = np.mean(sample)
    sample_means.append(sample_mean)
sample_means = np.array(sample_means)
# 18. Display first 10 sample means
print("\nFirst 10 Sample Means:")
for i, mean in enumerate(sample_means[:10], start=1):
    print(f'Sample {i}: {mean:.2f}')
# 19. Mean of all sample means
mean_sampling_distribution = sample_means.mean()
# 20. Standard deviation of sample means
std_sampling_distribution = sample_means.std(ddof=1)
print("\nSampling Distribution:")
print("Number of Samples:", num_samples)
print("Sample Size:", sample_size)
print("Mean of Sample Means:",
      round(mean_sampling_distribution, 2))
print("Standard Deviation of Sample Means:",

```

```

round(std_sampling_distribution, 2))
# 21. Histogram
plt.hist(sample_means, bins=10, edgecolor="black")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.title("Sampling Distribution of Sample Mean : Priti
Yadav T054")
plt.show()
# 22 & 23. Comparison
print("\nComparison:")
print("Original Sample Mean:",
      round(sample_mean, 2))
print("Mean of Sampling Distribution:",
      round(mean_sampling_distribution, 2))

```

```

# 25. Standard Error
standard_error = sample_std / np.sqrt(sample_size)
print("\nStandard Error:",
      round(standard_error, 2))
# 26. CLT explanation
print("\nCentral Limit Theorem:")
print("The sampling distribution of the sample mean")
print("becomes approximately normal as the sample size
increases.")
# 27. Population mean estimation
print("\nEstimated Population Mean:",
      round(sample_mean, 2), "hours")

```

OUTPUT:

First 10 Sample Means:

```

Sample 1: 2.25
Sample 2: 2.18
Sample 3: 2.45
Sample 4: 2.40
Sample 5: 2.32
Sample 6: 2.30
Sample 7: 2.42
Sample 8: 2.53
Sample 9: 2.27
Sample 10: 2.22

```

Sampling Distribution:

```

Number of Samples: 100
Sample Size: 30
Mean of Sample Means: 2.3
Standard Deviation of Sample Means: 0.13

```

Comparison:

```

Original Sample Mean: 2.28
Mean of Sampling Distribution: 2.3

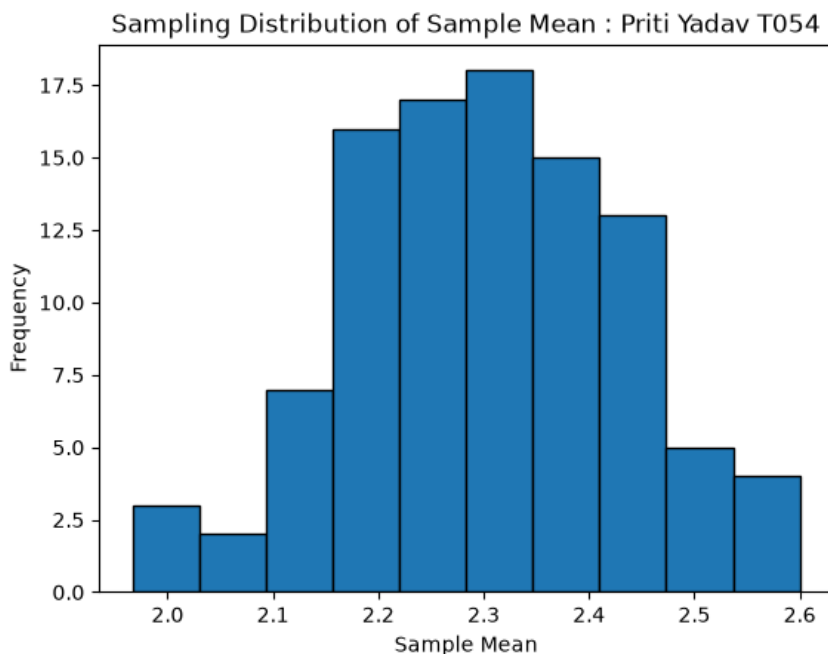
```

Standard Error: 0.18

Central Limit Theorem:

The sampling distribution of the sample mean becomes approximately normal as the sample size increases.

Estimated Population Mean: 2.28 hours



Conclusion:

The sampling distribution is centered close to the original sample mean. This demonstrates the Central Limit Theorem and shows that the sample mean can be used to estimate the population mean.